



ACCRA TECHNICAL UNIVERSITY

Department of Computer Science

BCS 404

Introduction to Data Science with Python

PROJECT WORK

From Deck Plans to Data: Exploring Survival on the Titanic

EDA · Statistics · Machine Learning

Dataset: Titanic Passenger Data (Kaggle)

Academic Year: 2025/2026 Second Semester

Lecturer: Dr. Joseph Dadzie

Student Name: Christian Ampeh Oduro

Index Number: 01252798B

Programme: BSc Computer Science

Date: July 2026

Table of Contents

1. Introduction
2. Dataset Description
3. Methodology
4. Results
5. Discussion
6. Conclusion
7. References
8. Appendix

1. Introduction

Data science combines statistical reasoning, programming and domain knowledge to extract insight from data. This project applies that workflow to passenger records from the RMS Titanic disaster of April 1912.

Aims: acquire and inspect data; clean missing/duplicate values; visualise patterns; run statistical analysis; train Logistic Regression for survival; discuss findings and limitations.

Software: Python, Jupyter Notebook, Pandas, NumPy, Matplotlib, Seaborn and Scikit-Learn.

2. Dataset Description

Source: Kaggle Titanic competition. File: train.csv. 891 passengers, 12 variables. Target: Survived.

3. Methodology

Cleaning: median Age; mode Embarked; drop Cabin; no duplicates. Predictors: Pclass, Sex, Age, SibSp, Parch, Fare, Embarked. Stratified 80/20 split. Logistic Regression (max_iter=1000).

4. Results

Survival ~38.4%. Strongest + correlation: SibSp-Parch (~0.42). Strongest - correlation: Pclass-Fare (~-0.55). Logistic Regression accuracy ~80.4%.

Chart titles below are written as questions to keep the story readable.

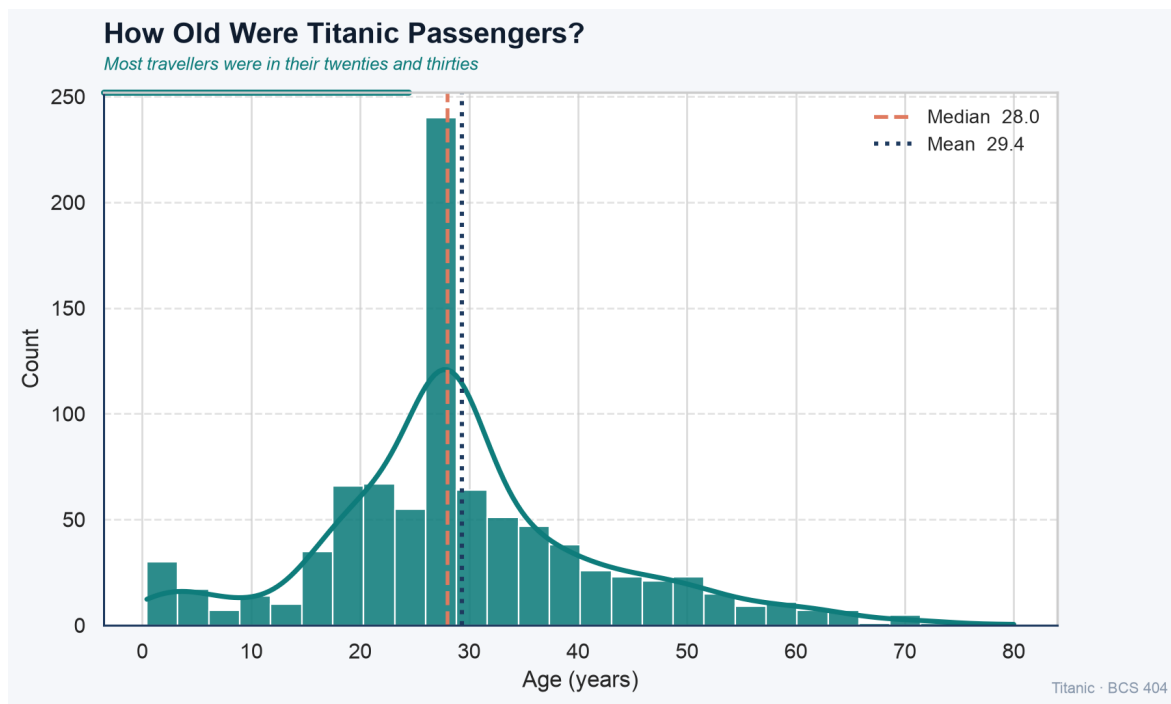


Figure 1. How old were Titanic passengers?

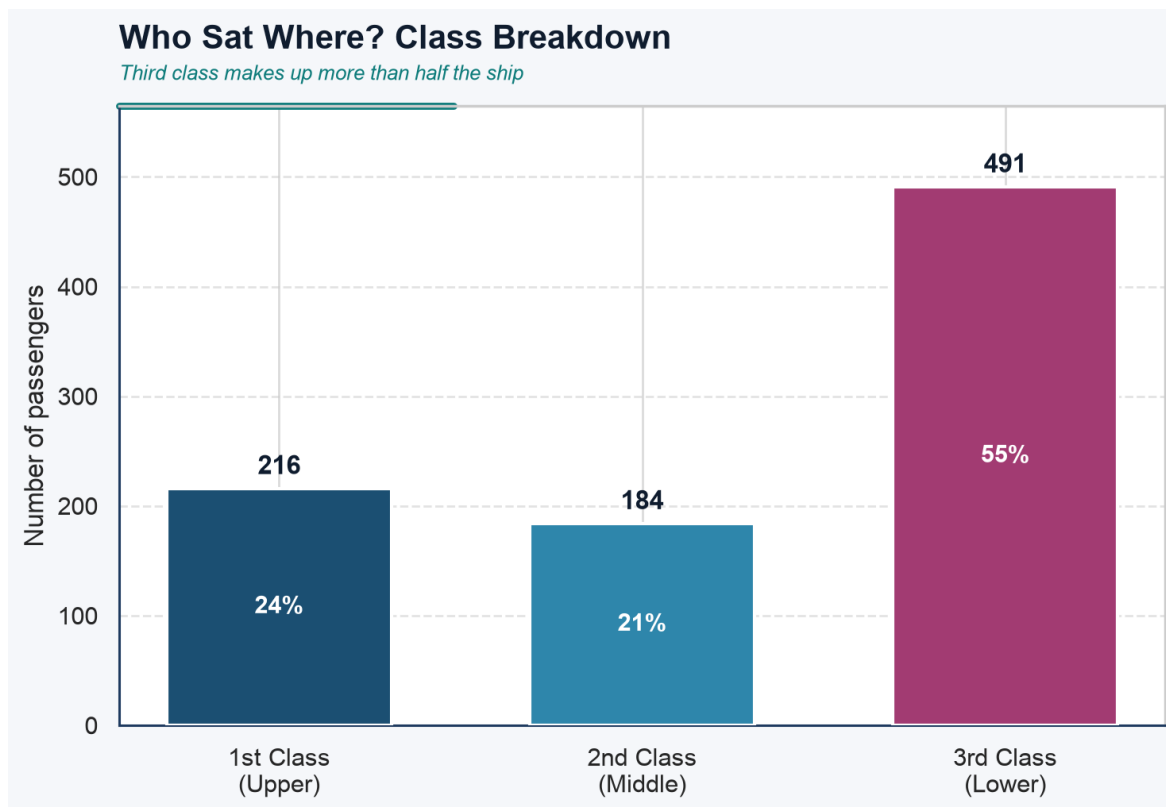


Figure 2. Who sat where? Class breakdown.

Younger in Steerage?

Median age falls from 1st class down to 3rd

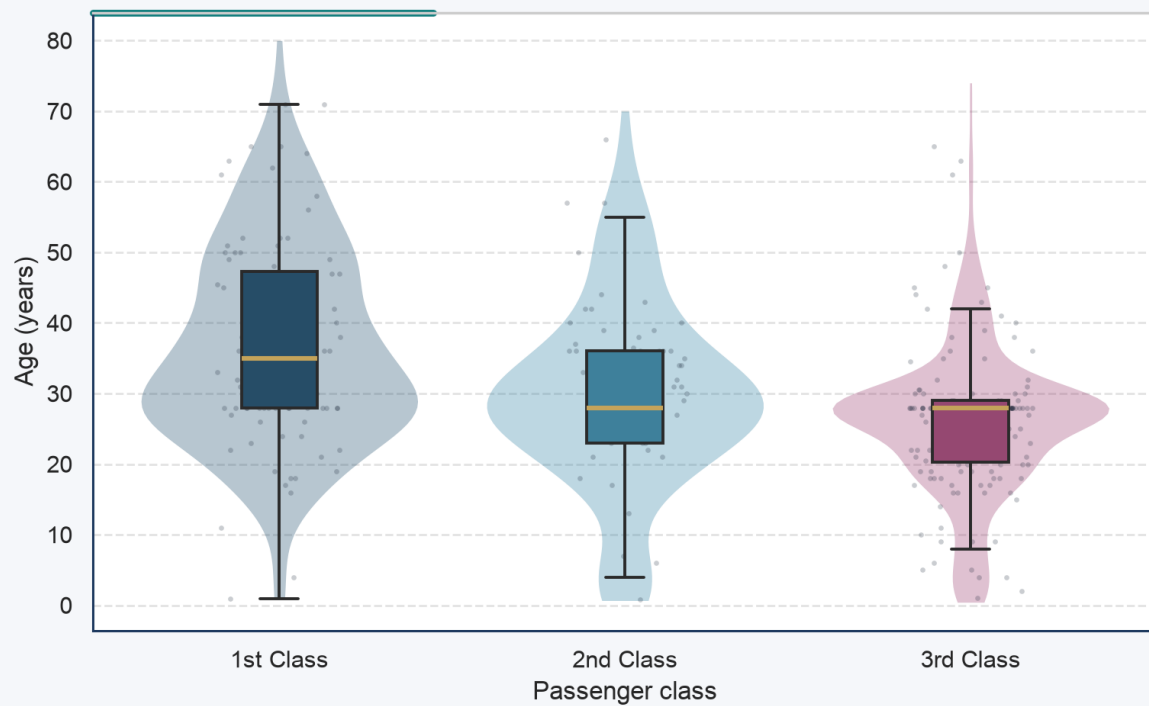


Figure 3. Younger in steerage? Age by class.

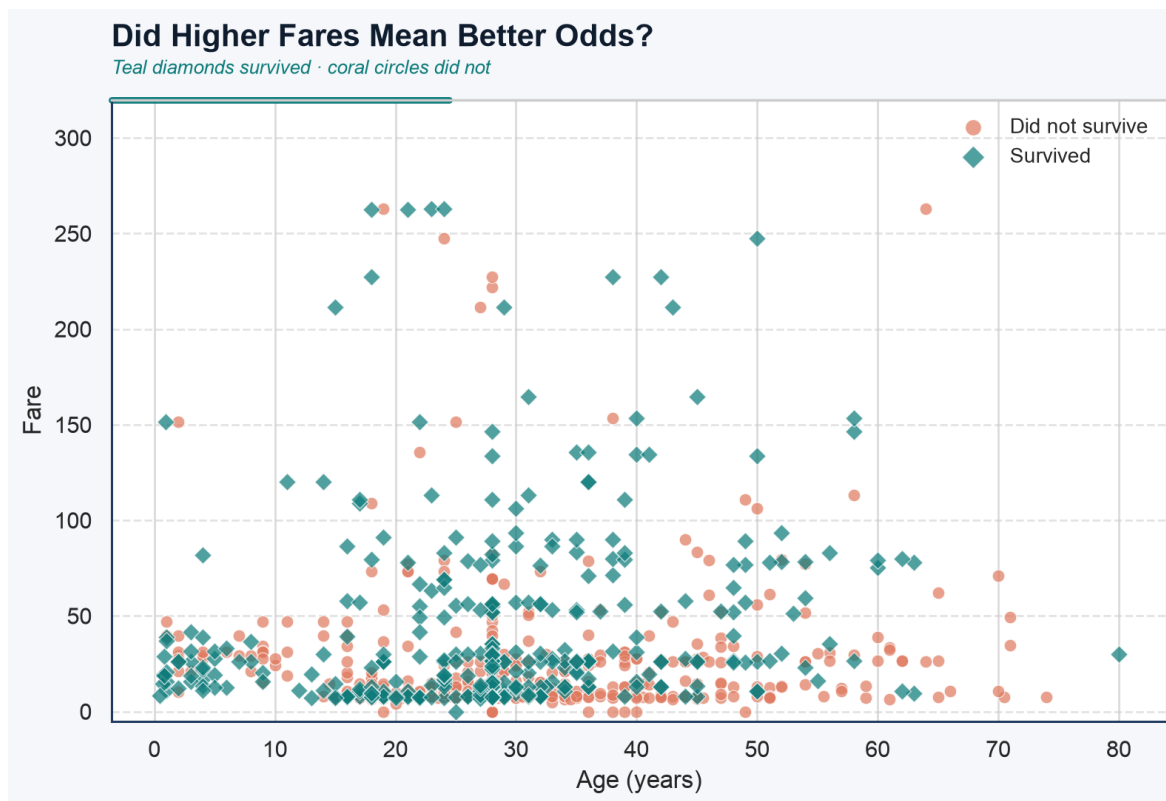


Figure 4. Did higher fares mean better odds?

What Moves Together?

Pearson correlations among numerical variables

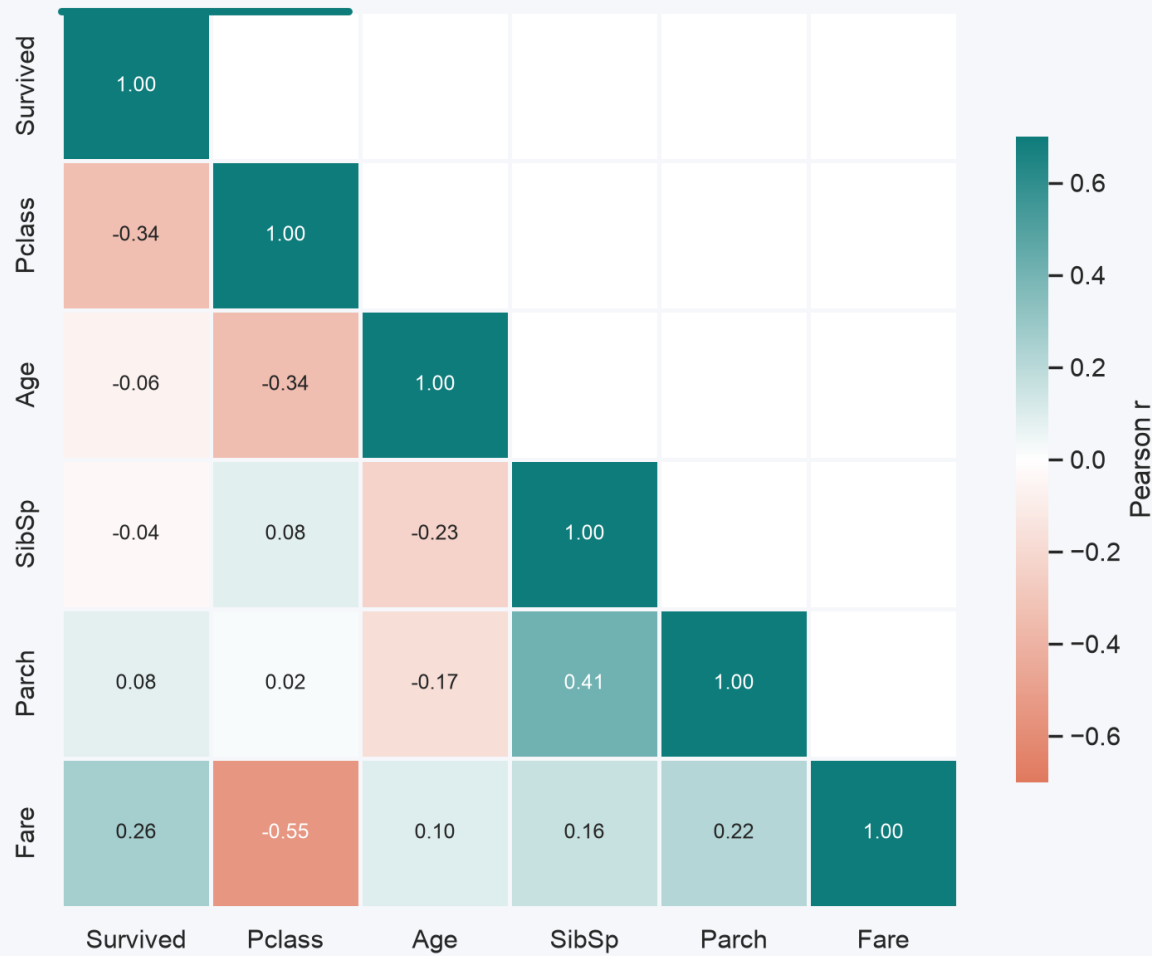


Figure 5. What moves together?

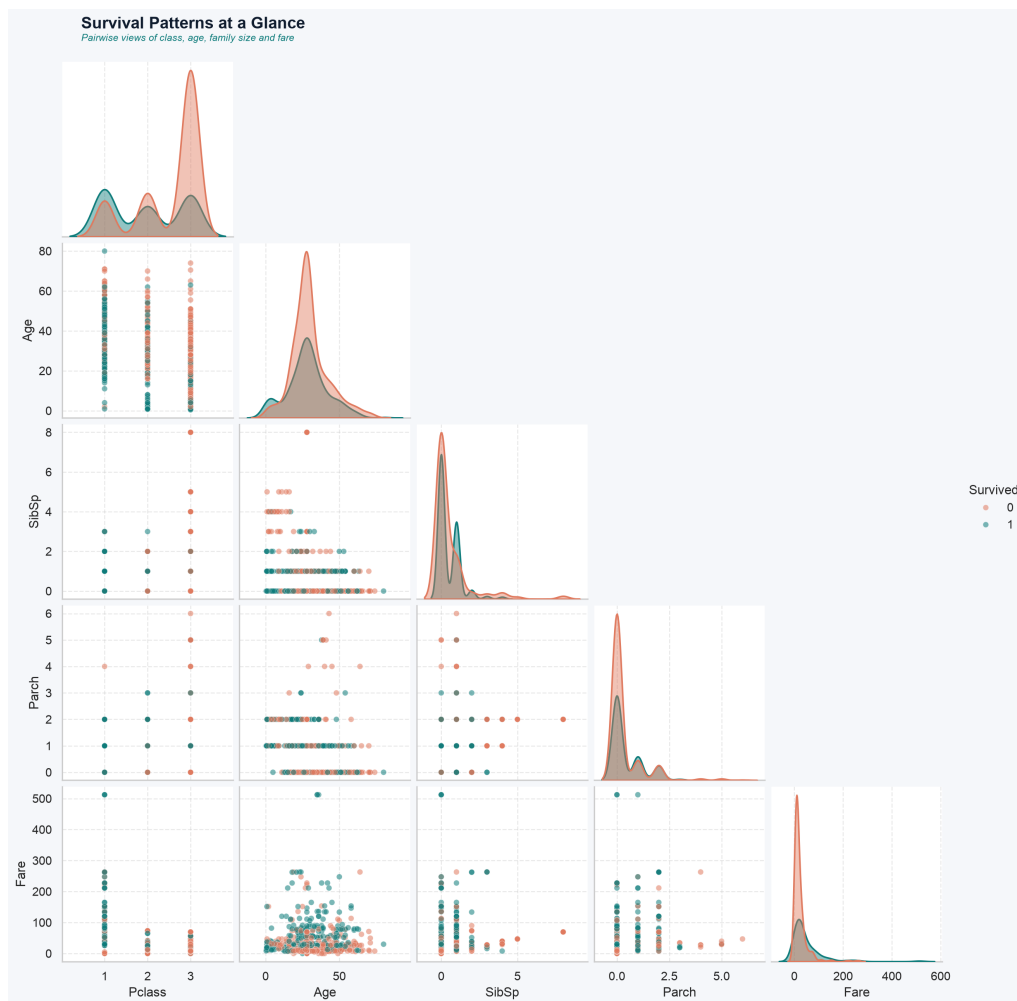


Figure 6. Survival patterns at a glance.

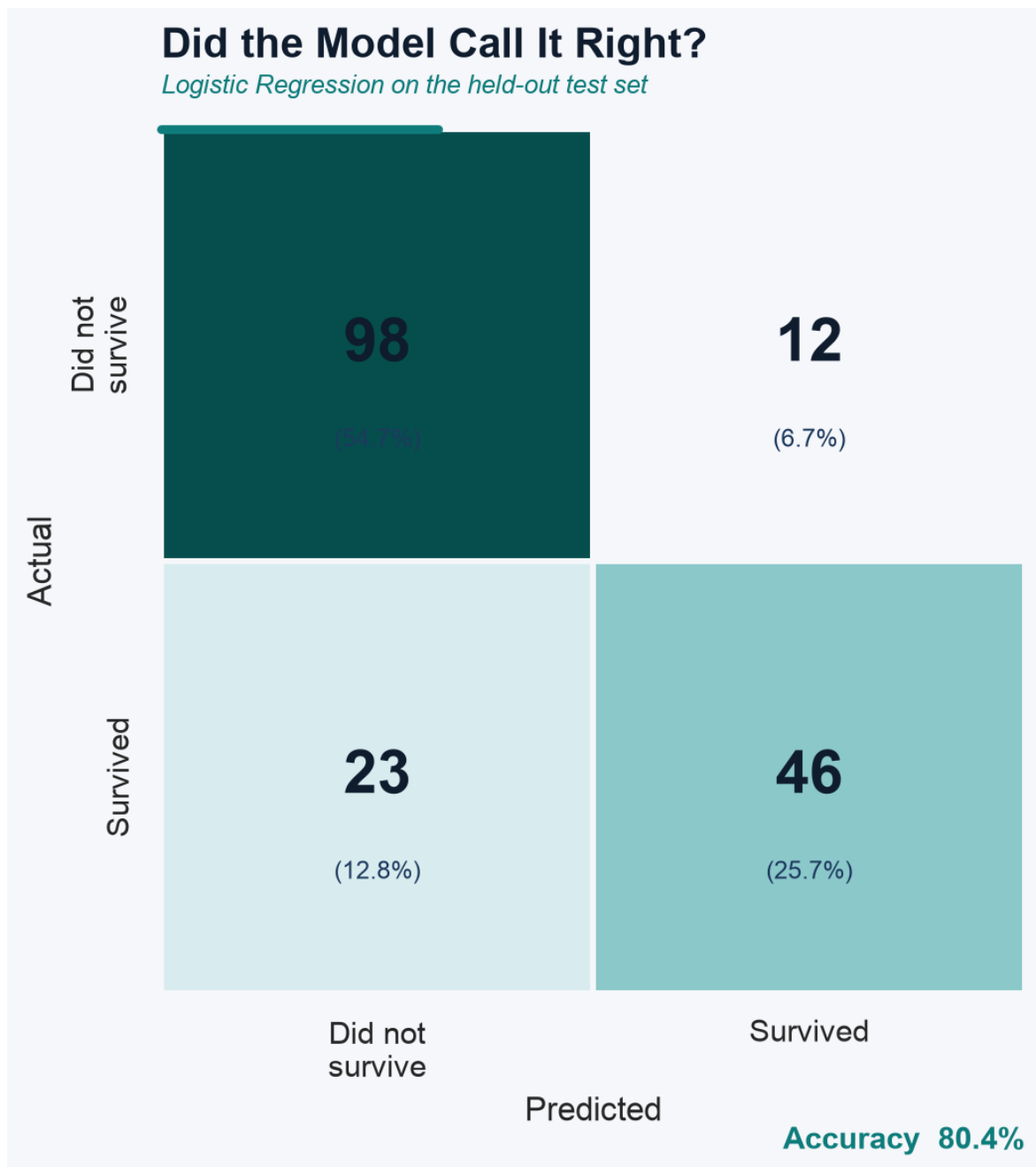


Figure 7. Did the model call it right?

5. Discussion

Major findings show sex and passenger class strongly shape survival. Statistical correlations support socio-economic inequality of outcomes. The model (~80% accuracy) beats the ~62% majority baseline, though survivor recall (~0.67) remains weaker than non-survivor recall.

Limitations: median age imputation; dropped Cabin; unused Name/Ticket titles; single algorithm and split.

Recommendations: engineer titles/family size; try ensemble models with CV; tune threshold for survivor recall.

6. Conclusion

End-to-end pipeline completed on Titanic data. Visual, statistical and ML results align: sex and class matter, and Logistic Regression provides an interpretable ~80% accurate baseline for BCS 404.

7. References

1. ATU BCS 404 Project Brief (2025/2026).
2. Kaggle Titanic competition data.
3. pandas, NumPy, Matplotlib, seaborn, scikit-learn documentation.

8. Appendix

Code: notebooks/BCS404_Titanic_EDA_ML.ipynb and src/titanic_analysis.py.

Prepared by Christian Ampeh Oduro (01252798B), Accra Technical University.